Intelligent Verification System for Concrete Compliance & QA Risk Management

BEAMP Cycle 6 Proposal — Ambient Systems Pte. Ltd.

Ivan Damnjanović, Founder & Lead Applicant

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About Ambient Systems

Global Presence

Founded in Singapore (2021) under NUS GRIP program. AI-driven compliance automation with operations spanning Singapore, Hong Kong, Canada, and Serbia.

WO for AI-powered building equipment compliance automation.

Multidisciplinary team combining civil engineering expertise, AI innovation, and cross-jurisdictional business development capabilities.

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Our Proven Expertise

AI/NLP Pipeline

Advanced multi-source data ingestion and processing capabilities

Compliance Automation

Specialized systems for building and equipment regulatory data

System Integrations

Seamless connectivity with PDFs, scans, WhatsApp, and supplier portals

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The Industry Challenge

Data Fragmentation

Concrete QA data scattered across paper delivery slips, PDFs, spreadsheets, and chat messages

Manual Processes

Time-intensive reconciliation leads to project delays, human errors, and compliance audit risks

Financial Risk

Finance teams approve milestone claims without comprehensive QA data verification

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The BEAMP Opportunity

Perfect Partnership

Expand Construction actively seeks automated concrete QA verification solutions for enhanced project oversight.

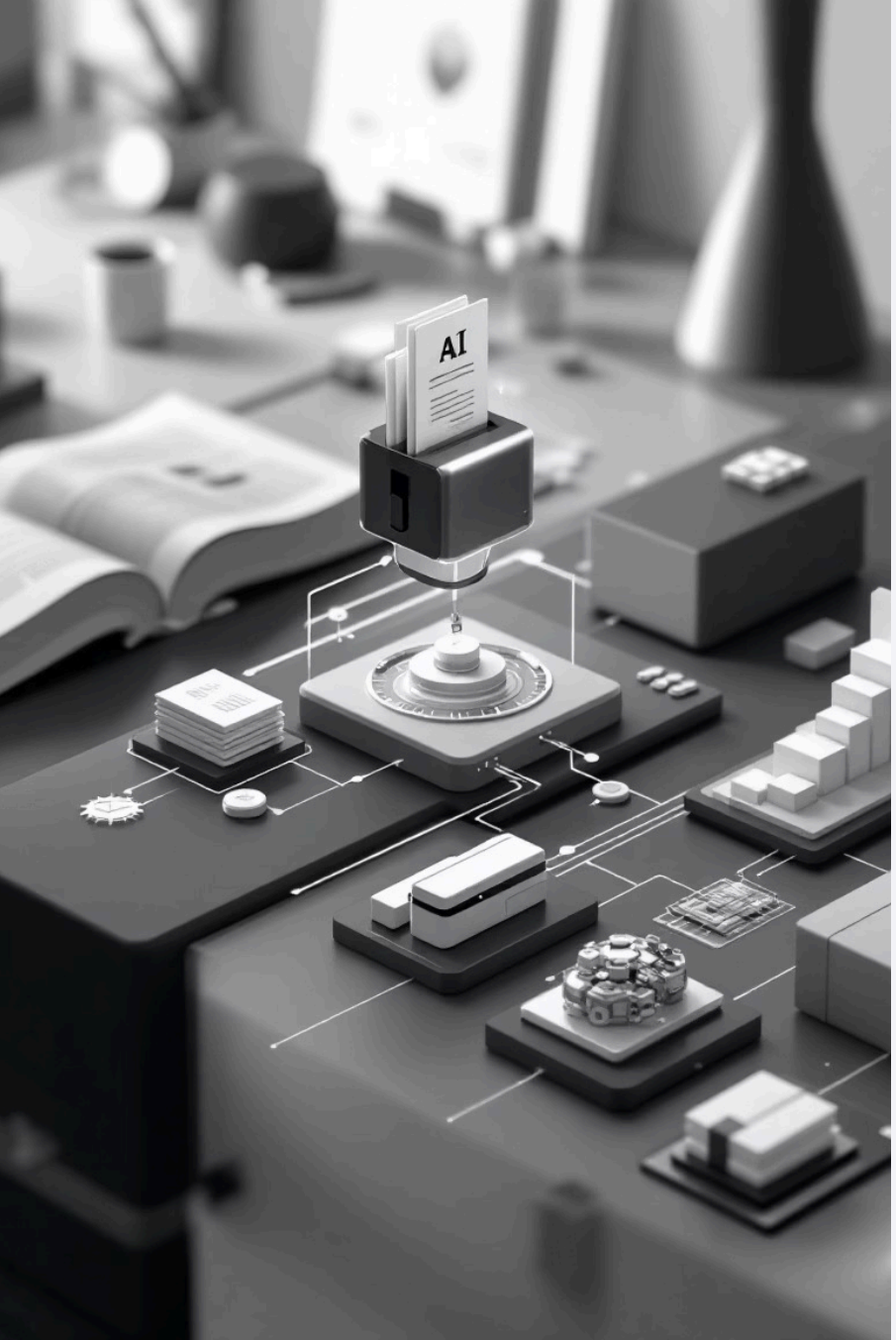
Strategic Benefits

BEAMP provides essential funding, real-world pilot environment, and ecosystem support for rapid development.



Ambient Systems uniquely positioned to deliver comprehensive AI-driven compliance automation.

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Our Solution Architecture

Multi-Source Ingestion

Automated data extraction from diverse document formats

AI Processing Engine

Intelligent reconciliation and anomaly detection

Interactive Dashboards

Real-time insights for QA, finance, and audit teams

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Core System Features



Universal Data Ingestion

Seamlessly processes PDFs, WhatsApp messages, electronic delivery orders, and spreadsheets from any source.



Intelligent Reconciliation

Automatically links delivery records, slump tests, and cube strength reports for complete traceability.



Smart Anomaly Detection

Configurable alerts flag missing data, non-conformance issues, and potential quality risks.



Role-Based Dashboards

Customized views for QA managers, finance teams, and audit compliance requirements.

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Pilot Implementation Plan

1

Months 1-2

Data Integration Phase

Deploy ingestion systems and develop reconciliation logic for Expand's workflows

2

Month 3

Dashboard Deployment

Launch interactive dashboards with live QA data visualization and reporting

3

Month 4

Validation & ROI

Performance measurement and system optimization based on real-world results

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Expected Impact & Outcomes

70%

Reduction in Manual QA

Eliminate time-intensive data collation processes

100%

End-to-End Verification

Complete traceability from delivery through testing and approval

24/7

Audit Readiness

Continuous compliance monitoring and milestone claim validation

Transform fragmented QA data into structured datasets supporting ESG reporting and regulatory compliance.

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Scalability & Future Vision

Technology Expansion

- Steel QA verification systems
- Prefabricated unit compliance
- BIM compliance automation

Business Model

SaaS platform with flexible pay-per-use API or enterprise deployment options.

> *"Turning fragmented QA data into real-time compliance intelligence"*

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